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INTEGRATION OF FLEXIBILITY, RELIABILITY AND MAINTENANCE IN PROCESS SYNTHESIS AND DESIGN

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ABSTRACT

In this paper the problem of simultaneously accounting for continuous uncertainty, equipment failures and maintenance policies during the design of processing systems is addressed. Based on analytical tools for quantifying flexibility, reliability and maintenance, a mixed-integer nonlinear programming model is proposed for the overall operability optimization of a process system at minimum cost. It is shown, through the design of a gas transmission network, that the integration of these stochastic operability measures at the early synthesis level results in inherently flexible and reliable process designs.

INTRODUCTION

The need for incorporating operability objectives in the design stage has been widely recognized (*Grossmann & Morari, 1981, Perkins, 1989, Morari, 1992*). While significant progress has been made in developing efficient methods for evaluating different operability characteristics (reliability, flexibility, controllability, availability) most approaches are limited to single operability citerion and/or confined to fixed design configurations. Only recently, the problem of simultaneously incorporating and optimizing more than one operability criteria during the design evolution has started to receive more attention (*Pistikopoulos et al., 1990, Straub & Grossmann, 1990, 1991, Mathisen & Skogestad, 1992*). Thomaidis and Pistikopoulos (1992 a,b) have introduced an integrated framework to simultaneously measure flexibility, reliability and maintenance of process design configurations; the framework is based on the following analytical tools:

- a *stochastic flexibility index, SF*, representing the maximum probability for a design to accommodate variations of continuous uncertain parameters (flowrates, stream compositions, demands, etc.).
- a *combined flexibility-reliability index, FR*, to measure the ability of a design to simultaneously cope with continuous process varaitions *and* equipment failures within a specified time horizon.
- a *combined flexibility-availability index, FA*, to assess the average performance of a processing system under continuous and discrete uncertainty by explicitly considering maintenance performed on the failed equipment. The effect of applying different maintenance policies on the process performance can be quantified and comparisons can be effectively performed.

This work focuses on the design optimization of these metrics at minimum cost; i.e. design decisions are based on the simultaneous consideration of economics and operability. A single parametric optimization model is proposed which properly explores the trade-offs by fully accounting for the continuous uncertainty, discrete equipment failures and maintenance policies.

The paper is organised as follows. First the optimization model is briefly introduced. Then, the design optimization procedure is discussed through its application to a gas transmission network problem, where the impact of considering the different operability criteria together with annualized cost is demonstrated.

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DESIGN OPTIMIZATION OF FLEXIBILITY, RELIABILITY AND MAINTENANCE

Consider the optimal design of a gas transmission network where the pipeline layout is fixed, but the actual number and optimal arrangement of the required compressors has to be determined. The different potential design alternatives are represented through the network superstructure shown in Figure 1a. Given are also : a process model together with cost data and specifications (for example, minimum pressure requirements at terminal points), a stochastic distributional model for the continuous uncertain parameters as well as reliability data and a set of different maintenance policies.

The problem is then to obtain the minimum cost design configuration with the optimal arrangement and the optimal capacity of the compressors as well as the optimal length and diameter of the pipeline segments, while maximizing the ability of the system to handle process variations and equipment failures.

Conceptually the problem can be posed as the design optimization of operability objectives subject to the process model process specifications and process economics as follows:

$$\max_{d,y} FR$$

where :

$$\left.\begin{aligned} FR &= \tfrac{1}{(T_b-T_a)} \int_{T_a}^{T_b} \textstyle\sum_{k=1}^{K} P^k(t) SF^k dt \\ P^k(t) &= \textstyle\prod_{i=1}^{n} \{r_i(t)^{x_i^k} (1-r_i(t))^{(1-x_i^k)}\} \end{aligned}\right\} \quad \text{combined Flexibility-Reliability}$$

$$\left.\begin{aligned} SF^k = & \int \cdots \int_{\{\theta : \psi(d^k,\theta) \le 0\}} J(\theta) d\theta \\ & k = 1, \cdots, K \end{aligned}\right\} \quad \text{Stochastic Flexibility} \qquad (P)$$

s.t.

$$\left.\begin{aligned} &\text{Process model} \\ &\text{Process specifications and constraints} \end{aligned}\right\} \quad f(y,d,z,\theta) \le 0$$

$$\text{Process economics} \qquad C(y,d,z) \le C^{max}$$

where: y is the set of binary variables defining the design configuration, d is the set of design variables (capacities, diameters, pipeline lengths), k, $(k = 1,..,K)$ is the set of operating states, whereas θ is the set of the continuous uncertain parameters; z denote the control variables of the process, x_i^k a binary variable to denote the operational condition of equipment i, $(i = 1,..,n)$ in state k (1 operable, 0 failed), $r_i(t)$ a reliability function of equipment i, $(T_a - T_b)$ a time horizon of interest, $P^k(t)$ denotes state probabilities; SF^k represents the stochastic flexibility index for state k, $J(\theta)$ is a joint probability function of the continuous uncertainty θ, and the expressions in $\psi(d^k,\theta) \le 0$ define the boundary of the feasible region in θ-space (*Pistikopoulos & Mazzuchi, 1990*), FR is the combined flexibility-reliability index for the specified time horizon $(T_a - T_b)$. f is a vector of inequality constraints and C is a scalar annualized cost function, whereas C^{max} represents a maximum value for the overall cost.

Note that problem (P) is a parametric in C^{max} mixed integer optimization problem which additionally involves integration over all feasible operating states; however, feasible states are different for different design configurations. To overcome this difficulty a Gaussian quadrature integration scheme is first employed with which analytical expressions of the combined flexibility-reliability index can be obtained as a function of the structural and design variables (y and d respectivelly) over all quadrature points. Then, a successive procedure is applied based on a modified Generalised Benders Decomposition scheme, where per iteration (fixed structure) a sequence of small optimization subproblems is solved for the feasible operating states (*Straub & Grossmann, 1993*). A parametric evaluation of the combined metric *vs.* C^{max} accounts for the systematic trade-off considerations between cost and operability.

The same methodology is also applicable for the design optimization of the combined flexibility-availability index when different maintenance policies are considered. If the failure and repair rates (μ, λ) are assumed constants and only two operating modes (failed or operable) are considered for each part of the equipment, then the state probabilities P^k can be evaluated based on a Markovian state-space model according to the number of the available service crews and the repair preference order of the equipment (*Thomaidis &*

Pistikopoulos, 1992b), as follows:

$$\left.\begin{array}{rl} FA = \sum_{k=1}^{K} P^k SF^k \quad where \quad \mathbf{P} = A^T\mu + B^T\lambda \\ \sum_{k=1}^{K} P^k = 1 \end{array}\right\} \text{combined Flexibility-Availability}$$

where: FA represents the combined flexibility-availability index and A, B are coefficients for the balance equations, when different maintenance policies are considered.

The approach will be further demonstrated through an example problem.

APPLICATION

The gas transmission network of Figure 1 is considered which involves one resource (A), two terminal points (C, D) and five potential positions for the installation of gas compressors. The location of the split point (B) is fixed. The diameters and lengths of the pipelines between the resource/demand points and the compressor/split points, together with the required capacity of the compressors are the design variables. Ideal gas behaviour is assumed and the gas temperature throughout the network is constant at 300 K. The process model and the required data for process economics and process specifications are shown in Table 1. The minimum required pressure at terminal points C and D, is specified at 700 and 500 psia respectively.

In order to establish a stochastic model for describing process variations, the inlet pressure and flowrate at source point A are considered as uncertain parameters, following normal distribution functions with mean values and variances given in Table 2. Equipment failure and repair rates are modeled via Weibull distribution functions, with parameters as shown in Table 2.

For the maximization of the combined flexibility-reliability index over a time period of 500 hrs, the time space is discretized in five points. For each operating state the same number of quadrature points per uncertain parameter dimension is also employed for the evaluation of the stochastic flexibility through the Gaussian quadrature integration scheme. For the maximization of the combined flexibility-availabiltiy index one service crew and an equipment service preference order of $D \rightarrow C \rightarrow 1 \rightarrow 2 \rightarrow 3$ have been specified here; an approach to also optimize the maintenance policy is discussed elsewhere (*Thomaidis & Pistikopoulos, 1993*). The stochastic metrics are then obtained parametrically as function of the overall cost. The optimal configuration consisting of three compressors is depicted in Figure 1.b, whereas the results of the detailed design are shown in Table 3. Trade-off curves between overall cost and different operability measures are also shown in Figure 2.

It is interesting to note that the exclusion of operability aspects from the design proceedure (case 3.4 Table 3) or even the inclusion of a single operability objective (i.e. reliability, R, or stochastic flexibility, SF, cases 3.1 and 3.2 of Table 3) results in severe overestimation of the system performance. This is clearly shown in Figure 2, where the design optimization of the stochastic flexibility index alone requires an overall cost of 3.7 millions (see case 3.1 of Table 3). Instead, when the combined flexibility-reliability index is maximized over a specified period of operation 500 hrs, the resulting optimal design for the same cost has an average performance of only 30%, case 3.2 of Table 3). If maintenance is performed on the failed parts of the equipment then the steady-state availability of the network increases up to 64% (again a lower value compared to the stochastic flexiblity index prediction). In order to achieve 90% steady-state network availability, the overall cost must be increased up to 4.1 millions (case 3.3 of Table 3). Note that the number of pumps remains the same, however the design values change significantly.

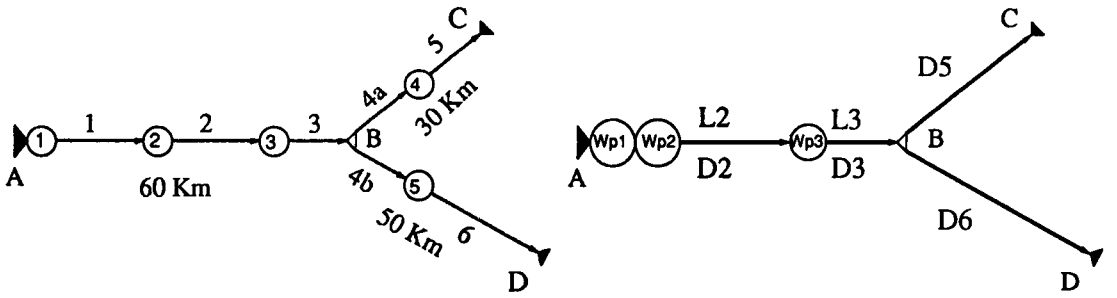


Figure 1: (a) Design superstructure model, (b) Optimal design configuration

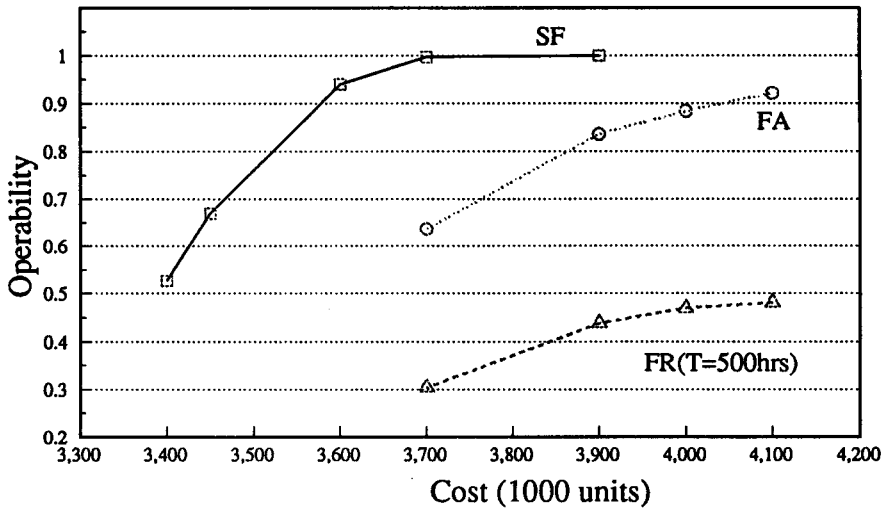


Figure 2: Trade-off curves

Table 1: The process model of the example

PROCESS MODEL					
Flow Equation :	$D_{ij}(P_i^2 - Pj^2)^\alpha = [s_g T(\frac{P_o}{k_1 T_o})^2]^\alpha (L_{ij}Q_{ij}^2)^\alpha$				
Compressor Work :	$Wp_j = (\frac{P_{dj}}{P_{sj}})^b [\frac{k_2 T \gamma}{(\gamma-1)n}] q_j$				
	$P_{dj} - P_{sj} \geq 0$		$W_{min} \leq Wp_j \leq W_{max}$		
	$P_{di} - P_{sj} \geq 0$		$W_{min} - TDR * W_{max} \geq 0$		
$(P_{sj},\ P_{dj})$:	suction & discharge pressure of the compressor at node j				
Q_ij :	the flowrate in pipeline segment connecting node i to node j				
q_j :	total flowrate of gas compressed by compressor at node j				
$(D_{ij},\ L_{ij})$:	diameter & length of pipeline segment connecting node i to node j				
s_g :	gas specific gravity	$(P_o,\ T_o)$:	standard conditions	T :	average gas temperature
$k_1,\ K_2$:	constants	γ :	heat capacity ratio	n :	compressor efficiency
α :	(3/16)	b :	$\frac{\gamma-1}{\gamma}$	TDR :	0.2

DESIGN SPECIFICATIONS & DATA			
max. capacity for the compressors		30,000 hp	
max. diameter for the pipelines		30 inch	
max. allowed pressure in the network		2000 psia	
fixed layout	L_{A-B} = 60 Km	L_{B-C} =30 Km	L_{B_D} =50 Km

Process Economics			
fixed compressor cost	**operating cost**	**maintenance cost**	**piping cost**
Cf = 100,000 $(compressor,\ yr)^{-1}$	Co = 10,000 $(hp\ yr)^{-1}$	Cm 3,000 = $(hp\ yr)^{-1}$	Cp = 870 $(Km\ inch\ yr)^{-1}$
cost function :	$\sum_i Cf_i + (Cm + Co)\sum_i Wp_i + Cp\sum_j (L_j D_j) \leq C^{max}$		

Table 2: Continuous & discrete uncertain parameters

continuous uncertain parameters			discrete uncertainty		
Inlet	Flowrate (MMSCF)	Pressure (psia)	Weibull : $f_i(t) = \alpha_i \beta_i t^{\beta_i - 1} e^{-\alpha_i t_i^\beta}$		
mean value	800	500	compressors	demand point 1	demand point 2
variance	63	53	$\alpha_{1,..,5} = 0.0001\ hr^{-1}$	$\alpha_C = 0.005\ hr^{-1}$	$\alpha_D = 0.006\ hr^{-1}$
			$\beta_{1,..,5} = 1.0$	$\beta_C = 1.0$	$\beta_D = 1.0$
		repair rates	$\mu_{1,..,5} = 0.05\ hr^{-1}$	$\mu_C = 0.83\ hr^{-1}$	$\mu_D = 0.125\ hr^{-1}$

CONCLUSIONS

In this paper a framework has been presented which enables the rigorous integration of operability objectives (flexibility, reliability and availability) with economics during the early design stages of process systems. A set of analytical tools have been presented for the systematic quantification of the different operability characteristics. These tools are integrated within a unified design framework for the simultaneous optimization of the operability characteristics of a processing system by (i) taking into account different maintenance strategies and (ii) exploring the trade-offs between investment, operating and maintenance cost and overall system performance.

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Table 3: Results

Cost $x10^{-3}$	Discrete state flexibilities					Pipeline : Diameters (in)				Lengths (Km)		Compressor Capacities (hp)		
(3.1)	maximization of the stochastic flexibility index, no equipment failures													
					SF	D_2	D_3	D_5	D_6	L_2	L_3	Wp1	Wp2	Wp3
3400					0.527	24.3	24.4	16.6	18.3	40.0	20.0	24,250	24,665	11,024
3450					0.670	24.3	24.4	16.8	18.5	39.5	20.5	25,125	25,558	11,354
3600					0.941	25.4	25.4	17.4	19.1	38.5	21.5	28,000	28,470	11,722
3700					0.997	26.1	26.1	17.8	19.6	38.0	22.0	29,250	29,740	12,100
(3.2)	maximization of the combined flexibility-reliability index FR operating period (0-500 hrs), without maintenance													
	R	FR	SF^1	SF^2	SF^3	D_2	D_3	D_5	D_6	L_2	L_3	Wp1	Wp2	Wp3
state probabilities			0.177	0.174	0.129									
3700	0.481	0.303	0.910	0.820	-	24.9	24.9	22.6	20.2	33.0	27.0	26,237	27,153	9,397
3900	0.481	0.438	0.995	0.885	0.852	25.6	25.6	22.0	23.4	45.1	14.9	27,034	27,960	11,723
4000	0.481	0.470	1.000	0.973	0.962	26.0	26.1	22.4	23.9	44.4	15.6	28,748	29,706	12,329
4100	0.481	0.481	1.000	1.000	1.000	26.6	26.6	22.7	24.7	43.2	16.8	30,000	30,000	13,769
(3.3)	maximization of the combined flexibility-availability index FA one service crew, service preference order : $D \rightarrow C \rightarrow 1 \rightarrow 2 \rightarrow 3$													
		FA	SF^1	SF^2	SF^3	D_2	D_3	D_5	D_6	L_2	L_3	Wp1	Wp2	Wp3
state probabilities			0.836	0.061	0.032									
3700		0.636	0.761	0.001	-	25.2	25.2	18.8	20.1	34.2	25.8	25.130	26,019	3,895
3900		0.836	0.999	0.002	-	26.2	26.3	18.8	20.2	25.2	34.8	28,900	29,880	7,250
4000		0.884	0.991	0.899	-	26.5	26.5	22.2	20.1	30.7	29.3	29,010	29,983	7,478
4100		0.922	1.000	0.911	0.793	26.8	26.8	22.4	23.2	40.1	19.9	27,347	28,307	6.732
(3.4)	cost minimization without consideration for operability objectives													
						D_2	D_3	D_5	D_6	L_2	L_3	Wp1	Wp2	Wp3
3352						23.9	23.9	16.5	17.8	38.3	21.7	23,852	24,704	10,588